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Use of electronic cigarettes and asthma across the lifespan

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Abbreviations:

CDC= Centers for Disease Control and Prevention

COPD= chronic obstructive pulmonary disease

COVID-19= coronavirus disease 2019

ENDs= electronic nicotine-delivery systems

EVALI= e-cigarette or vaping product use-associated lung injury

FDA= Food and Drug Administration

FeNO= exhaled nitric oxide

- 47 FEV₁= forced expiratory volume in one second
- 48 FVC= forced vital capacity
- 49 IgE= Immunoglobulin E
- 50 NIOSH= National Institute of Occupational Safety and Health
- 51 PEF= peak expiratory flow
- 52 PG= propylene glycol
- 53 PMTA= Premarket Tobacco Product Applications
- 54 SHS= second-hand smoke
- 55 THC= tetrahydrocannabinol
- 56 VG= vegetable glycerin
- 57 U.S.= United States

ABSTRACT

In the last 15 years, the tobacco industry has marketed electronic cigarettes (e-cigarettes) as a tool to help adults to quit cigarette smoking, while also promoting their use in children and adolescents. As expected, this led to a marked increase in e-cigarette use (also called vaping) among children and adolescents in the U.S. Although the prevalence of vaping decreased after the 2019 EVALI epidemic and stricter regulations fueled by concurrent adoption of cigarette and marijuana smoking in youths, approximately 10% of middle- and high-school students in the U.S. report vaping. A growing body of experimental and epidemiologic evidence has implicated e-cigarette use in the pathogenesis of respiratory diseases such as chronic obstructive pulmonary disease (COPD), lung cancer, interstitial lung disease, and asthma. In this narrative review, we first evaluate the types and composition of e-cigarettes and trends in e-cigarette usage and then assess findings from experimental and epidemiologic studies of e-cigarette exposure and asthma published to date. We end by discussing future directions for research in this field while emphasizing the need for prevention of and screening for e-cigarette use in youths and adults, particularly those affected with asthma.

INTRODUCTION

Cigarette smoking is a major cause of mortality among adults worldwide, accounting for more than 480,000 deaths per year in the United States (U.S.) alone. Second-hand smoke (SHS) exposure *in utero* and in early life increases the risk of incident asthma during childhood^{1, 2} and SHS worsens disease control or severity in children with asthma^{3, 4 5-7}.

Over the last 15 years, the tobacco industry has marketed electronic cigarettes (e-cigarettes) as a tool to help adults quit cigarette smoking, while also promoting their use in children. Not surprisingly, this led to a marked increase in e-cigarette use (also called vaping) among high school students in the U.S., from 1.5% in 2011⁸ to a peak of 27.5% in 2019⁹. Following the epidemic of E-cigarette or Vaping Product Use-associated Lung Injury (EVALI) in 2019 and growing concerns about higher risk of initiation and concurrent use of cigarette smoking and marijuana use in youths using e-cigarettes, these products became more strictly regulated (e.g., by mandating a minimum purchase age of 21 years) and their use decreased among middle- and high-school students, with ~10% reporting e-cigarette use in 2023¹⁰.

A growing body of experimental and epidemiologic evidence has recently implicated e-cigarette use in the pathogenesis of respiratory diseases such as chronic obstructive pulmonary disease (COPD), lung cancer, interstitial lung disease, and asthma. In this article, we first discuss the types and composition of e-cigarettes and trends in e-cigarette usage, review findings from experimental and epidemiologic studies of e-

cigarette exposure and asthma published to date, and discuss future directions for research in this field while also emphasizing the need to prevent and screen for e-cigarette use in youths, particularly those affected with asthma.

Literature review

We searched PubMed, Medline, Scopus, and the Cochrane Database for all studies reporting on e-cigarette use and asthma that were published in English until December 2025, using the following keywords: (“e-cigarette”, “e-cig”, “vaping”, “vapes”, “electronic nicotine delivery systems”, “ENDs”, “JUULing”, or “Pods”) and asthma (“asthma”, “exacerbations”, “asthma outcomes”, “asthma controls”, “emergency department visit”, “hospitalization”, or “lung function”). Abstracts and full-texts were reviewed by K.P. and Y.Y.H. and excluded if they were: review articles, articles not focused on use of e-cigarettes/vaping, or reports of studies focusing on conditions other than asthma (e.g., cystic fibrosis, COPD, EVALI, cancer, or COVID-19).

E-cigarettes versus combustible cigarettes

E-cigarettes were introduced into the U.S. in the mid-2000s, initially marketed as a safer and less addictive alternative to combustible cigarettes, and as a means for people to stop smoking conventional cigarettes^{11, 12}. E-cigarettes have many names, including vapes, vape pens, e-hookahs, mods, or electronic nicotine-delivery systems (ENDs), and are sometimes referred to by brand name, such as “JUULing”¹³. All such devices commonly share the same basic anatomy: a battery-powered heating device which heats a liquid (hereafter referred to as e-liquid), turning the liquid into an aerosol

(referring to both gases and particles produced during atomization, often incorrectly called “vapor”), which is then inhaled by the user¹¹. Batteries are most often lithium, which carry risks of explosion and injury^{14, 15}.

Since their introduction, e-cigarette devices have undergone changes and can be categorized as first-, second-, third-, or fourth-generation devices, although e-cigs continue to change and many devices have overlapping features¹⁶. First-generation devices typically resemble a traditional combustible cigarette and are small and disposable, while second-generation devices (often referred to as “vape pens”) are larger and frequently feature a liquid tank and separate battery¹⁶. Third-generation devices (a.k.a. mods) typically offer the user variable features such as adjustable wattage, voltage, and/or power and a refillable tank¹⁶. These devices led to the identification of higher amounts of toxins being found in e-cigarette aerosols when e-liquids are heated to higher temperatures or higher Wattage is used. Fourth-generation (a.k.a. pods) often resemble every-day items, are small and easy to hide, and often have prefilled or refillable cartridges or are disposable¹⁶. Device generation is important when considering potential harms, as the materials used to make devices are typically plastics and metals, which end up potentially being inhaled as well. Detailing if the device is used inhaling from a regular e-cigarette versus dripping (i.e. dropping e-liquid onto a hot coil and breathing in the aerosol) is important, as dripping utilizes higher temperatures, leading to the generation of greater amounts of toxins¹⁷.

In addition to variable devices, e-liquid demonstrates marked variability in contents. Most e-liquids have three basic components (nicotine, humectant, and flavoring¹⁷⁻¹⁹), differing in chemical composition and/or strength. For example, earlier devices often used freebase nicotine, a volatile product, at concentrations of 1-6 mg/ml. When an e-cigarette aerosol is inhaled by a user of such devices, the nicotine is more likely to off-gas from the aerosol droplets and deposit in the mouth/upper respiratory tract, where it is absorbed into the blood. Absorption in the oral cavity/upper respiratory tract is slower than that with conventional cigarettes, resembling that for nicotine replacement therapy^{20, 21}. In 2015, nicotine salts (which have a vastly higher nicotine concentration of 59 mg/ml and are less harsh and bitter to the user) were introduced. Nicotine salts are less volatile than freebase nicotine thus having greater potential to reach alveolar spaces^{20, 21}. Further, the amount of nicotine a user inhales depends on the device generation and the experience of the user²².

In 2023, the industry introduced several e-cigarette and smokeless products, including those containing 6-methyl nicotine, a synthetic nicotine analog that is pharmacologically more potent than nicotine. In late 2023 and early 2024, the major U.S. e-cigarette suppliers Nicotine River and ECBlend introduced 'PMTA-exempt' products with the brand names 'Nixamide' or 'Nixodine' or 'Nixotine', with nicotinamide as the main active ingredient. While the Food and Drug Administration (FDA) requires tobacco product manufacturers to submit Premarket Tobacco Product Applications (PMTA) for new products (granting marketing approval only if deemed appropriate for the protection of

public health), these new products were marketed as 'PMTA-exempt', representing a significant regulatory challenge for the FDA²³.

The humectants used in e-liquid are typically a combination of glycerol (commonly called vegetable glycerin; VG) and propylene glycol (PG). Different ratios of PG:VG change the mouthfeel and look of e-cigarette aerosols. Most e-liquids used for first, second and third generation devices used higher ratios of PG:VG, such as 80:20, 70:30 and 50:50, while fourth generation devices tend to use higher concentrations of VG, such as 20:80 and 30:70 PG:VG. A myriad chemical flavorings (e.g., volatile organic compounds such as aldehydes, alcohols, esters, monoterpenes) added to e-liquids have concentrations capable of exceeding National Institute of Occupational Safety and Health (NIOSH) regulated levels¹⁸. For example, Tierney et al. found concentrations of vanillin at 33 mg/mL, with potential inhalation of 105 to 165 mg/day representing twice the recommended NIOSH limits¹⁸. Importantly, new toxicants generated or released during the heating process should also be considered when evaluating health harms²⁴. For example, formaldehyde has been found in e-cig aerosol (but not in the e-liquid) generated from heating propylene glycol to high temperatures²⁵, and metals have also been found in aerosols generated from the type and temperature of the coil²⁴. Depending on the type of device, users could also inhale tetrahydrocannabinol (THC), synthetic cannabinoids (Spice or K2), opioids (fentanyl and heroin), methamphetamine, and cocaine. Such variance in devices, e-liquid chemical concentration and composition, voltage/heat, type of coil(s) and user experience pose challenges for

researchers and clinicians in determining the health harms of e-cigarettes and providing cessation guidance and counseling.

Trends in e-Cigarette Use

As previously mentioned, there was a marked increase in e-cigarette use among adolescents in the U.S. from 2011 to 2019, with a subsequent decrement following stricter regulations. However, the U.S. Centers for Disease Control and Prevention (CDC) Foundation reported an increase in total e-cigarette sales from the beginning of 2020 to the end of 2023, with the highest increase observed for use of non-tobacco flavored e-cigarettes in young adults²⁶.

A 2024 U.S.-based internet survey found that e-cigarettes were the most commonly used tobacco product among middle- and high-school students, with a preference for flavored e-cigarettes²⁷. E-cigarettes were the most common currently used tobacco product (5.9%) among all students, followed by nicotine pouches (1.8%), combustible cigarettes (1.4%), cigars (1.2%), smokeless tobacco (1.2%), other oral nicotine products (1.2%), heated tobacco products (0.8%), hookahs (0.7%), and pipe tobacco (0.5%)²⁷.

While there was a slight decrease in e-cigarette use among high-school students in 2024 relative to 2023¹⁰, there was no change in the frequency of use among middle-school students. In a separate cross-sectional study in Canada, England and the U.S., there was a shift in the type of device used for vaping between 2017 and 2023, with disposables becoming the leading type of e-cigarette used by youths of all socio-demographic groups, overtaking refillable tank e-cigarettes and pods/cartridges²⁸.

Prevention of vaping in youth remains crucial, as e-cigarette use among adolescents and young adults has also been associated with use of other substances (alcohol and marijuana) over time^{29, 30}.

Use of e-cigarettes has been increasing among adults worldwide. From 2020 to 2023, e-cigarette use increased in U.S. adults, particularly in men, non-Hispanic Black and White persons, and Hispanic persons³¹. While adults aged 18–24 years were most likely to use e-cigarettes, younger e-cigarette users were more likely to be never smokers than older users³². Adult e-cigarette users are more likely to use other substances such as marijuana, with young adults having the highest prevalence of marijuana vaping^{33, 34}. Similar trends were seen in the European Union, where data collected from the 2014 - 2017 Adult Special Eurobarometer for Tobacco Survey found that 1.8% of adults were current regular e-cigarette users in 2017, up from 1.5% in 2014³⁵. In a study conducted in England between 2016-2023, the rising popularity of disposable vapes and historic declines in nicotine use had reversed, with nicotine use rising primarily by sharp increases in vaping among young people³⁶.

Second-hand exposure and asthma

In children, SHS exposure can occur trans-placentally while in utero or through cohabitating with active tobacco smokers³⁷. While such exposure types are the same for e-cigarette aerosols, traditional cigarettes create a side-stream or continuous release of chemicals when burning, while second-hand e-cig exposure is a result of the user exhaling the aerosol¹⁷. Third-hand exposure refers to nicotine and/or other chemicals

within inhalants that are deposited on surfaces, such as carpets, curtains and furniture, and can result in exposure from dermal, ingestion, and/or inhalational (i.e. off-gas) routes^{37, 38}.

Prenatal second-hand exposure

Table 1 summarizes murine studies of prenatal exposure to e-cigarettes and experimental asthma. In rats, pre- and post-natal exposure to nicotine (given subcutaneously) has transgenerational effects on asthma, resulting in inherited asthma for up to three generations³⁹. In mouse models, exposure to e-cigarette aerosols during pregnancy has been linked to dysregulation of gene expression, restricted alveolar growth, and alteration of lung development in fetal lungs, as well as airway inflammation in mothers and their offspring, with DNA methylation changes in the offspring^{40, 41}. In other murine models, in utero exposure to nicotine-containing e-cigarette aerosols (vanilla and mint flavored JUULs), followed by house dust mite allergen challenge later in life, led to decreased tidal volume, increased neutrophils in bronchoalveolar lavage and increased peribronchial and perivascular eosinophils, and decreased methylation of the promoter region of *Il-10ra* and abnormal expression of Arginase 1 (all associated with increased risk of allergic airway inflammatory disease or experimental asthma)^{42, 43}.

Limited evidence from human studies suggests that perinatal e-cigarette exposure negatively impacts newborn craniofacial and pulmonary development and can result in stillbirth and life-threatening consequences⁴⁴. To date, no human study has examined the relation between pre- or peri-natal e-cigarette exposure and childhood asthma.

Postnatal secondary exposure

Few studies have examined secondary exposure to e-cigarette aerosols and asthma in young children (**Table 2**). Among children with asthma aged 4 to 14 years who participated in the School Inner-City Asthma Study⁴⁵, those whose caregivers reported any cigarette/e-cigarette exposure in the household had higher urinary cotinine (a biomarker of exposure to nicotine vapes and cigarette smoke) and asthma symptom days and increased odds of hospitalizations for asthma compared with those whose caregivers reported no cigarette or e-cigarette use in the child's household. However, those findings were likely confounded by SHS from conventional cigarettes, as only one participant was solely exposed to e-cigarette aerosols. Among 54 children aged 5 to 17 years who participated in a retrospective study in Messina (Italy), there was no significant association between second-hand exposure to e-cigarette aerosols at home and number of asthma symptom days or need for rescue medication⁴⁶.

E-cigarette use and asthma in adolescents

The use of e-cigarettes is rising and, in some cases, exceeds traditional combustible cigarette use among middle and high school students⁴⁷. A few cross-sectional and prospective studies have examined primary and secondary exposure to e-cigarette vapor in adolescents and young adults and asthma outcomes (**Table 3**).

In a cross-sectional study of 1,345 students aged 16 to 19 years in Kuwait, those who never smoked cigarettes but used e-cigarettes had significantly higher odds of current wheeze and current asthma than those who never smoked or vaped⁴⁸. Moreover, dual

users of e-cigarettes and conventional cigarettes were more likely to have uncontrolled asthma symptoms⁴⁸. Similar results for asthma and asthma symptoms or asthma control were reported in other cross-sectional studies of adolescents in South Korea and the U.S.⁴⁹⁻⁵³. In some of those studies, e-cigarette use was associated with asthma in adolescents independently of cigarette smoking^{48-51, 53} or marijuana use^{50, 53}. In another large cross-sectional study in the U.S., second-hand exposure to END was associated with 27% increased odds of asthma exacerbations in the prior year, even after adjusting for SHS exposure and use of cigarettes, cigars, hookah, or e-cigarettes⁵⁴.

In a prospective study conducted between 2014 to 2019 in 2,090 high school students in Southern California, participants who reported second-hand nicotine vape exposure had higher odds of bronchitic symptoms and shortness of breath than nonexposed participants, even after adjusting for age, sex, race or ethnicity, parental education, SHS exposure, second-hand cannabis exposure, and primary vaping or smoking⁵⁵. In the same cohort, current sole users of e-cigarettes had higher odds of wheeze, bronchitic symptoms, and shortness of breath than never users⁵⁶. On the other hand, exclusive use of e-cigarettes was not significantly associated with age of onset of asthma or newly diagnosed asthma in three prospective studies of U.S. adolescents⁵⁷⁻⁵⁹.

E-cigarette use and asthma in adults

Several cross-sectional and prospective studies have examined primary exposure to e-cigarette aerosols in adults and asthma outcomes (**Table 4**).

In a prospective study of primary use of ENDS in the prior 30 days (recent use) and asthma onset among adolescents and adults, recent use of ENDS was associated with earlier onset of asthma in adults⁵⁹. Similar results were reported in a cross-sectional study of adults in France, in which e-cigarette use was associated with asthma and chronic bronchitis, even among participants who had never used conventional cigarettes or marijuana⁶⁰. Likewise, a study of U.S. adults showed that e-cigarette use was associated with increased odds of current asthma and any respiratory symptoms or illness compared with never use of e-cigarettes⁶¹⁻⁶⁴, with older adults and ever users of other tobacco products having greater odds of having any respiratory illness than younger adults and those who never used other tobacco products⁶³. In a study of 50 adult smokers with (n=25) and without (n=25, controls) asthma, participants with asthma -but not controls- had significantly reduced FEV₁/FVC and peak expiratory flow (PEF) and significantly increased fraction of exhaled nitric oxide (FeNO) 15 minutes after e-cigarette use⁶⁵.

In contrast to the studies reviewed above, others have reported no significant association between ENDS use and asthma morbidity in adults^{66, 67}.

DISCUSSION

E-cigarette use has grown rapidly in the US, particularly among adolescents and young adults. Despite the popularity of these products, there is a paucity of well-designed studies of their acute and long-term effects on asthma.

Limited evidence suggests a potential link between second-hand exposure to aerosols from e-cigarettes and worse asthma symptoms or control, particularly in adolescents. While the negative findings of one study of mostly prepuberal children can be explained by very limited statistical power⁴⁵, three large studies of adolescents (two cross-sectional and one prospective) reported significant associations between second-hand exposure to aerosols from e-cigarettes and current asthma, asthma symptoms or asthma attacks, even after accounting for active cigarette smoking and SHS exposure^{48, 54, 55}. However, all studies examined asthma or asthma outcomes through questionnaires and lacked lung function measures or validated scales of asthma control.

Current evidence from seven cross-sectional studies⁴⁸⁻⁵⁴ and one prospective study⁵⁶ suggests an association between e-cigarette use and current (active) asthma, asthma symptoms and worse asthma control in adolescents, with some studies suggesting synergistic detrimental effects of e-cigarette use and cigarette or cannabis smoking on asthma outcomes^{48, 52}. On the other hand, findings from two prospective studies^{57, 59} do not support an association between e-cigarette use and incident (de novo) asthma or earlier onset of asthma in adolescents. All published studies relied solely on questionnaires for assessment of asthma and asthma outcomes.

Findings from the largest four^{59-61, 64} of six^{59-61, 64, 66, 67} cross-sectional studies published to date suggest a link between e-cigarette use and current asthma or asthma symptoms in adults, with one of these studies reporting a potential dose-response relationship⁶¹.

While all these studies relied solely on questionnaires, a single controlled study of exposure to e-cigarettes showed negative effects of e-cigarette use on lung function measures and increased FeNO in 25 adult smokers with asthma⁶⁵.

Interpretation of most published studies of e-cigarette use and asthma is limited by their cross-sectional design (which precludes determining temporal relationships); non-comprehensive assessment of asthma, asthma severity, and asthma-related phenotypes or endotypes, often relying exclusively on self-reported questionnaire data without assessing lung function or biomarkers in blood and exhaled breath (e.g., blood eosinophils, FeNO, total and allergen-specific IgEs). Similarly, most studies have relied on self-report of e-cigarette use and use of tobacco and other substances (e.g., cannabis) without obtaining quantitative measures of exposure to assess dose-response relationships, while some studies have failed to account for concurrent active or passive exposure to tobacco products and other substances. Moreover, most published studies were conducted in the U.S., limiting their generalizability to populations living in other geographic regions.

Large prospective studies with long duration of follow up are needed to establish whether e-cigarette use or its known complications (e.g., EVALI) leads to new onset of

asthma and to determine short- and long-lasting effects on asthma outcomes in children and adults. Such studies should include careful assessment of e-cigarette use (including frequency, duration, number of inhalations/puffs, new and emerging devices, and types of flavorings and additives), concurrent exposure to tobacco products and other substances, and other environmental risk factors (e.g., outdoor pollutants) while examining potential effects on well-characterized asthma phenotypes and endotypes by including measures of lung function, biomarkers in blood and exhaled breath, and indicators of asthma severity or control). Randomized cessation trials in adolescents and adults who have asthma and use e-cigarettes are needed to support evidence-based interventions.

Figure 1 shows proposed mechanisms for potential effects of e-cigarette exposure (primary or secondary) on asthma, based on results of experimental studies. E-cigarette exposure may cause or worsen asthma through abnormal lung development and structure, alterations of T helper (Th)2 and non-Th2 immune responses, increased oxidative stress, airway inflammation and hyperresponsiveness, impaired mucociliary clearance and enhanced susceptibility to viral infections, and altered epigenetic programming^{41-43, 68-70}. In parallel with human studies, future experimental studies should further elucidate causal mechanisms and address long-term effects of e-cigarette exposure and co-exposure with other tobacco products, and toxic effects of e-liquid components and flavoring agents.

Ongoing and future research endeavors should not preclude vigorous efforts to develop and promote policies to prevent e-cigarette use in children and young adults, including regulatory actions against marketing of e-cigarette products to children and adolescents, enforcing restricted access to e-cigarettes by minors, educational programs targeting parents of middle- and high-school children at risk, and enhanced efforts to develop vaping cessation strategies, especially for adolescents and young adults. Such efforts are justified by reported potential detrimental effects on the respiratory health of persons with asthma⁷¹ (including EVALI⁷² and eosinophilic pneumonitis^{73, 74}) and increased risk of concurrent smoking of cigarettes and marijuana^{33, 75}.

Further, clinicians caring for patients with asthma should advise against and screen for e-cigarette use and use of tobacco products and marijuana in children and adolescents, using visual aids for ease of exposition⁷⁷. For adults trying to quit using combustible cigarettes, strategies such as behavioral treatment, cognitive therapy, or pharmacotherapies approved by the FDA should be recommended instead of e-cigarettes.

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FIGURE LEGEND

E-cigarette use may cause or worsen asthma by altering lung development and structure, increasing airway Inflammation, and altering gene expression through epigenetic changes.

Table 1. Prenatal exposure to e-cigarettes and respiratory outcomes in animal experimental studies

Reference	Animal model	Exposure(s): device type and key constituents	Respiratory outcome(s)/me- chanisms	Key findings and study limitations
Epigenetic mechanism				
Rehan et al., 2013 ³⁹	Male and female offspring of time-mated, first-timed pregnant, pair-fed Sprague Dawley rat dams (F ₀).	F ₀ received placebo (diluent) or nicotine (1mg/kg sc) in 100 µl volume daily from embryonic days 6 of gestation to postnatal day (P)21. F ₁ , F ₂ and F ₃ were not exposed to nicotine	In F ₁ (n=12) and F ₃ (n=13) offspring at P21: • Pulmonary function • Tracheal tension • Myofibroblast differentiation	Compared with the placebo group, perinatal nicotine exposure caused: • ↑ lung resistance and ↓ dynamic compliance of the lung following methacholine challenge in F ₁ and F ₃ male and female rats • ↑ tracheal constriction in response to acetylcholine in male rats, but not in female rats • ↑ fibronectin mRNA and protein levels and ↓PPARγ mRNA and protein levels in cultured fibroblasts from F ₁ and F ₃ rats <u>Limitations:</u> - Sex-specific effects of perinatal nicotine exposure on myofibroblast differentiation could not be examined - Did not measure AHR
Chen et al., 2018 ⁴⁰	Male offspring of 12-week-old female BALB/c mice	<i>In vivo</i> whole-body air chamber exposure of room air (sham); e-cig liquid without nicotine (E-cig0); or e-cig liquid with 18mg/ml nicotine ([50/50 PG/G]) with tobacco flavor additives (E-cig18) by a device of NEBBOX twice 15 minutes daily from 6 weeks before gestation to P20	At P1, P20, and 13-weeks: • Physical growth • Lung development • Lung inflammation • Epigenetic remodeling	Compared with sham group: • At P20, E-cig18 offspring had ↓body weight; both E-cig0 and E-cig18 offspring had bigger retroperitoneal fat mass • At P20, <i>PDGF</i> mRNA levels were upregulated in E-cig0 and E-cig18 offspring • At 13 weeks, ↓IL-1β and ↑TNF-α were found in E-cig0 and E-cig18 offspring and ↑IL-6 in E-cig0 offspring • At P1, global methylation ↑ more than 3-fold in E-cig0 offspring and ↑ more than 2-fold in E-cig18 offspring <u>Limitations:</u> - Sex-specific effects were not studied - Lack of mechanistic studies - Non-testing of different flavors or range of concentrations

Orzabal et al., 2022 ⁴¹	Male offspring of 8- to 12-week-old timed pregnant pair-fed Sprague Dawley rat dams	<i>In vivo</i> whole-body air chamber exposure of room air (controls); e-cig aerosol vaping without nicotine (EC-Base); e-cig aerosol vaping containing 10% nicotine (80/20 PG/G) by a Zeus sub ohm e-cig atomizer (EC-Nic) for 3-hr/day, 5 days per week from GD5-20	At GD21 (fetal), P4 and P10 (neonatal): • Physical growth • Lung transcriptome • Lung development	Compared with controls: • ↓ body weight and length in EC-Nic group at GD21 and P4 • Dysregulated expression of genes in fetal lungs: encoding protease inhibitors and inflammatory pathways in EC-Nic group and encoding tryptases and fibroblast proliferation in EC-Base group • ↓ fixed lung weight in EC-Nic group at P4 • ↑ free space within lung parenchyma and ↓ alveolar separation within the lung in EC-Nic group at P4 • ↑ total respiratory system resistance and ↓ static compliance for the pulmonary system in EC-Nic group at P10 <u>Limitations:</u> - Sex-specific effects were not studied - The exposure was not continuous and didn't span preconception or post parturition - The e-cigs did not contain flavorings, which is uncommon for commercial use
Cahill et al., 2022 ⁴²	Male and female offspring of 12-week-old female BALB/c mice	<i>In vivo</i> whole-body air chamber exposure of mint-flavored JUUL aerosol containing 5% nicotine salt (with PG, G, and benzoic acid) or HEPA-filtered air for 1-hr/day (controls) for 20 consecutive days during gestation	At birth and 11 weeks: • Physical growth • Lung development • Airway inflammation • Genetic and epigenetic changes	Compared with controls, mint-flavored JUUL aerosol exposure: • ↓ body weight and length at birth • ↓ airspace fraction and ↑ tissue fraction in the lung indicating delayed progression of lung development • Upregulated genes associated with Th2 inflammation, allergy, and asthma in the lung at birth and at 11-week-old male and female offspring • ↑ neutrophils in BALF of male and female adult offspring with HDM-induced asthma • ↓ methylation of the promoter region of <i>IL10ra</i> in adult offspring with HDM-induced asthma <u>Limitations:</u> - May not generalize to different types of e-cigarettes or ENDS device, flavors or nicotine concentrations

Noël et al., 2023 ⁴³	Male offspring of 12-week-old female BALB/c mice	<i>In vivo</i> whole-body air chamber exposure to vanilla-flavored e-cig aerosol containing 18 mg/ml nicotine (50/50 PG/G) by a device of eVic VTC mini mod or filtered air for 2-hr/day (controls) for 20 consecutive days during gestation	At birth or 7 weeks: • Lung transcriptome • Lung function • BALF differentiation	Compared with controls, vanilla-flavored e-cig aerosol exposure: • Dysregulated ≥ 65 genes in female and male offspring at birth, including genes encoding T-cell immune responses and chemokine signaling, extracellular membrane proteins, and oxidative stress responses • $\downarrow$ tidal volume and presence of lung mucous metaplasia in 7-week-old male mice with HDM-induced asthma • $\uparrow$ neutrophils and $\downarrow$ macrophages in BALF, $\uparrow$ peribronchial and perivascular eosinophils, and augmented alveolar polymorphonuclear cells in 7-week-old male mice with HDM-induced asthma <u>Limitations:</u> - Inability to distinguish key ingredients driving the responses - Cannot generalize to various flavors or mixtures of e-cig use - Gene expression was measured only at birth - Only male offspring were exposed to allergens
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AHR=airway hyperresponsiveness, BALF= bronchoalveolar lavage fluid, CRL=crown-rump length, e-cig=e-cigarette, END=electronic nicotine delivery systems, G=glycerin, GD=generational day, HDM=house dust mite, HEPA= high-efficiency particulate air, IL=interleukin, PG=propylene glycol, P=postnatal day, PDGF= platelet-derived growth factor, PPAR γ =peroxisome proliferator-activated receptor gamma, Th=T-helper, TNF- α = tumor necrosis factor-alpha,

Table 2. Postnatal exposure to e-cigarettes and asthma-related outcomes in children

Reference	Study population	Exposure(s):	Asthma-related Outcome(s)	Key findings and study limitations
Cross-sectional study				
Ruran et al., 2022 ⁴⁵	133 students aged 4-14 years with active asthma, recruited between 2015 to 2020 (School Inner-City Asthma Intervention Study), U.S.	Urinary level of cotinine (biomarker for environmental tobacco smoke exposure) and NNAL (4-methylnitrosamino-1-3-pyridyl-1-butanol, biomarker for tobacco smoke)	Parental or caregiver's report of: • Number of asthma symptom days over 2 weeks • Presence and number of hospitalizations for asthma in the previous year • Composite asthma severity index (CASI)	Adjusting for age, sex, race/ethnicity, income, obesity, use of controllers, prematurity, family history of asthma, data collection month, and urine specific gravity: • Each log₁₀ increase in cotinine was associated with increased maximum symptom days ($\beta=0.16$, 95% CI= -0.02- 0.34) and ≥ 1 hospitalization for asthma in the previous year (aOR=1.35, 95% CI=1.06-1.72). • Each log₁₀ increase in NNAL was associated with ≥ 1 hospitalization in the previous year (aOR=2.72, 95% CI=1.29, 5.71). • In a stratified analysis, similar results were reported in children with allergic sensitization, but no significant associations were reported in non-sensitized children • Urinary level of cotinine or NNAL was not associated with CASI <u>Limitations:</u> - Inability to assess temporal relationships - Only one student had e-cig exposure without dual use - Specific effects from e-cigs cannot be determined
Costantino et al., 2024 ⁴⁶	54 children aged 5-17 years with asthma diagnosis, recruited in 2023, Messina, Italy	Exposure to e-cig aerosol in the last year at home vs. non-exposure (control group)	Parental or caregiver's report of: • Number of asthma exacerbations in the previous year • Need for rescue therapy and/or therapeutic step-up • ACT/cACT	No significant differences between the exposed and unexposed groups in bivariate analyses of: • Number of days with asthma symptoms ($p=0.27$) • Need for rescue ($p=0.19$) or step-up ($p=0.3$) therapy • Asthma control: ACT ($p=0.3$)/ cACT score ($p=0.4$) <u>Limitations:</u> - Inability to assess temporal relationships - Small sample size, no multivariable analysis - Potential information and recall bias for self-reported data
ACT=asthma control test, aOR= adjusted odds ratio, cACT=children asthma control test, CI=confidence interval, e-cig=electronic cigarettes				

Table 3. Use e-cigarettes and asthma or asthma-related outcomes in adolescents

Reference	Study population	Exposure(s):	Asthma-related Outcome(s)	Key findings and study limitations
Cross-sectional studies				
Alnajem et al., 2020 ⁴⁸	1,345 adolescents aged 16-19 years recruited between January and May of 2019, Kuwait	• Never, former, or current (in the past 30 days) use of e-cig and/or cigarette smoking • Never used e-cigs or cigarettes (control) • Current (in the past 7 days) exposure to SHA from e-cig and SHS in the home or public vs. unexposed (control)	Self-report of: • Current wheezing • Current asthma • Current uncontrolled asthma symptoms	Adjusting for sex, age, exposure to household SHS and SHA, and exposure to SHS and SHA in public: • Current sole e-cig use with no history of cigarette smoking was associated with current wheeze (aPR= 1.54, 95% CI= 1.01-2.45) and current asthma (aPR= 1.85, 95% CI= 1.03-3.41) • Current dual use of e-cig and cigarettes was associated with current wheeze (aPR= 1.87, 95% CI= 1.44, 2.42), current asthma (aPR, 95% CI= 1.33-2.76) and uncontrolled asthma (aPR=1.97, 95% CI= 1.35-2.88) Adjusting sex, age, cigarette smoking, exposure to household SHS, and exposure to SHS or SHA in public: • Frequent exposure (≥ 3 days) to household SHA was associated with current wheeze (aPR=1.30, 95% CI= 1.04-1.59), current asthma (aPR=1.56, 95% CI= 1.13-2.16), and current uncontrolled asthma symptoms (aPR=1.88, 95% CI= 1.35-2.62) compared with those with no SHA exposure <u>Limitations:</u> - Inability to examine temporal relationships - Potential information and recall bias - Lack of data on adherence to asthma medications in participants with asthma - No information on duration and frequency of household SHA exposure

Cho et al., 2016 ⁴⁹	35,904 students in 10 th -12 th grades in (mean age=16.4 years) recruited in 2014, South Korea	Current (in the past 30 days) e-cig use vs. never use (control)	Self-report of: - Physician-diagnosed asthma in the past 12 months - Severe asthma: ≥ 4 days of school absences due to asthma in the past 12 months	Adjusting for gender, city size, multi-cultural family status, overweight, SHS at home, history of dermatitis and allergic rhinitis: - In never cigarette smokers: current e-cig use was associated with physician-diagnosed asthma (aOR=2.74, 95% CI= 1.30-5.78) and severe asthma (aOR= 15.42, 95% CI= 5.11-46.57) <u>Limitations:</u> - Inability to assess temporal relationships - Potential recall bias for self-reported exposures and outcomes - Data on use of asthma medications and healthcare utilization were not available
Schweitzer et al., 2017 ⁵⁰	6,089 adolescents in 9 th -12 th grades (mean age=15.8 years) recruited in 2015, Hawaii, U.S.	Current (in the past 30 days) e-cig use vs. never use (control)	Self-report of: - Ever asthma - Current asthma	Adjusting for age, gender, race/ethnicity, education intentions, overweight, cigarette smoking, and use of marijuana: - Current e-cig use was associated with ever asthma (aOR=1.48, 95% CI= 1.24- 1.78) and current asthma (aOR= 1.20, 95% CI= 1.00-1.44) <u>Limitations:</u> - Inability to assess temporal relationships - Lack of clinical assessment or medical records for asthma - Lack of information on potential confounders such as residential contexts and family hardship
Roh et al., 2023 ⁵¹	3,042 high school students in Texas and 32,885 high school students in the U.S., aged 13-17 years and recruited for the 2015-2019 YRBS	Ever and current (in the past 30 days) e-cig use vs. never use (control)	Self-report of ever asthma	Adjusting for age, sex, race/ethnicity, BMI, other substance use, and depression: - Among adolescents who had never used combustible products (cigarettes, cigars, and marijuana), ever e-cig use was associated with asthma among adolescents from Texas (aOR = 1.32; 95% CI = 1.06-1.66) and from the U.S. (aOR = 1.18, 95% CI = 1.02-1.37) compared with never users - This association was not reported in those ever-used combustible products <u>Limitations:</u> - Neither temporal nor dose-response relationships could be examined

				- Potential recall bias - Lack of information on socioeconomic status
Williams et al., 2023 ⁵²	150,634 high school students in the 10 th and 12 th grades recruited in 2019–2020 for the California Student Tobacco Survey, U.S.	Ever • Use of e-cigs (with nicotine and/or flavorings) • Dual use of e-cigs and cigarettes • Dual use of e-cigs and cannabis • Triple use of e-cigs, cigarettes, and cannabis • Never used any products (control)	Self-report of: • Lifetime asthma (asthma, but not in the past year) • Recent asthma: (ever had asthma and had an asthma attack in the past year)	Adjusting for age, sex, parental education, race/ethnicity, and household use of e-cig, cigarette or cannabis: • Ever use e-cig only was associated with lifetime asthma (aOR= 1.10, 95% CI= 1.02-1.18) but not with recent asthma (aOR= 1.12, 95% CI= 0.97-1.28) • Dual use of e-cig and cigarettes was associated with decreased lifetime asthma (aOR= 0.71, 95% CI= 0.52-0.96) • Dual use of e-cig and cannabis was associated with lifetime asthma (aOR= 1.17, 95% CI= 1.12-1.23) and recent asthma (aOR= 1.09, 95% CI= 1.00-1.19) • Triple use of e-cig, cigarettes, and cannabis was associated with lifetime asthma (aOR= 1.14, 95% CI= 1.06-1.24) and recent asthma (aOR= 1.19, 95% CI= 1.03-1.37) <u>Limitations:</u> - Inability to assess temporal relationships - Lack of data on age of asthma onset and frequency of asthma attacks
Wills et al., 2020 ⁵³	14,765 adolescents in the 9 th -12 th grades (aged 14-18 years) in the 2017 US YRBS	Ever and current (in the past 30 days) e-cig use vs. never use (control)	• Self-report of ever asthma	Adjusting for age, gender, overweight, obesity, race/ethnicity, use of cigarettes, and use of marijuana: • Ever use of e-cig (aOR= 1.15, 95% CI= 1.02-1.30) and current use of e-cigarettes (aOR= 1.30, 95% CI= 1.10-1.53) was associated with ever asthma • Current e-cig use and cigarette smoking had additive estimate effects on ever asthma (aOR for dual use=1.62, 95% CI= 1.32-1.99) <u>Limitations:</u> - Inability to assess temporal relationships - No data on current asthma

				- Lack of information on type of e-cig device or household SHS exposure
Bayly et al., 2019 ⁵⁴	11,830 children aged 11-17 years with self-reported asthma from the 2016 Florida Youth Tobacco Survey, U.S.	Current (in the past 30 days) secondhand exposure to ENDs aerosols in a room and/or in a car vs. unexposed (control)	Self-reported asthma attacks in the previous 12 months	Adjusting for age, sex, race/ethnicity, metropolitan status, housing, SHS exposure and use of cigarettes, cigars, hookah, or e-cig: • Secondhand exposure to ENDs aerosols was associated with ≥ 1 asthma attack in the previous 12 months (aOR=1.27, 95% CI =1.11-1.47) <u>Limitations:</u> - Inability to assess temporal relationships - Potential recall bias and misclassification of the outcome - Lack of information on whether family member switched to ENDs due to an asthma attack in the child - Limited generalizability
Prospective studies				
Islam et al., 2022 ⁵⁵	2,090 adolescents in the 11 th and 12 th grades (mean age 17.3 years) in the CHS from 2014 to 2019 (Waves 1-5), Southern California, U.S.	Current (in the past 30 days) secondhand exposure to nicotine vape in the household vs. unexposed (control)	Self-report of respiratory symptoms in the previous year: • Bronchitic symptoms • Wheeze • Shortness of breath	Adjusting for study wave, age, sex, race, parental education, primary smoking or vaping of nicotine or cannabis, and exposure to secondary smoking or vaping/smoking cannabis: • Secondhand exposure to nicotine vaping in the household was associated with increased bronchitic symptoms (aOR=1.40, 95% CI=1.06, 1.84) and shortness of breath (aOR=1.53, 95% CI= 1.06, 2.21), but not with wheeze (aOR=1.21, 95% CI=0.84, 1.72) • Stronger association were observed in participants who were neither smokers or vapers <u>Limitations:</u> - Potential measurement errors and misclassification of the exposure and respiratory outcomes - Potential recall bias

Tackett et al., 2024 ⁵⁶	2,094 adolescents in the 11 th and 12 th grades (mean age= 17.3 years) enrolled in the CHS from 2014 to 2018 (Waves 1-4), Southern California, U.S.	Current e-cig use (in the past 30 days) vs. never user (control)	Self-report of respiratory symptoms in the previous year: • Bronchitic symptoms • Wheeze • Shortness of breath	Adjusting for study wave, age, sex, race, lifetime asthma diagnosis, parental education, concurrent use of cigarettes, cannabis, and secondhand exposure to e-cig, cigarettes, and cannabis: • E-cig use was associated with wheeze (aOR=1.41, 95 % CI=0.99, 2.01), bronchitic symptoms (aOR=1.55, 95 % CI=1.18, 2.05), and shortness of breath (aOR= 1.48, 95 % CI=1.01, 2.18). • These associations remained significant among children without self-reported asthma <u>Limitations:</u> - Non-quantification of e-cig use (i.e. frequency, number of inhalations, duration) - Potential recall bias - Lack of information on type of device, e-liquid flavors, and nicotine types
Patel A, et al., 2024 ⁵⁷	9,141 adolescents aged 12-17 years in the U.S. PATH study from 2013 to 2019 (Waves 1-5)	Current (in the past 30 days): • Exclusive use of ENDS • Exclusive use of cigarettes • Dual use of ENDS and cigarettes • Never or non-current user (control)	Self-reported or parentally reported incident diagnosed asthma	Adjusting for age, sex, race/ethnicity, BMI, parental education, urban/rural setting, SHS exposure, and household combustible tobacco use: • Current use of ENDS exclusively (aHR=1.25, 95% CI= 0.77–2.04) or in combination with cigarettes (aHR=1.54, 95% CI= 0.92–2.57) was not associated with incident diagnosed asthma • Exclusive cigarette use was associated with incident diagnosed asthma (aHR=1.61, 95% CI=1.18-2.20) <u>Limitations:</u> - Potential recall bias - Limited sample size for exclusive ENDS use and dual use - Lack of data on known risk factors for asthma such as allergy, air pollution, or family history of asthma - Potential selection bias due to differential loss to follow-up

Yao et al., 2024 ⁵⁸	11,148 students aged 12-17 years from Waves 4.5 and 5 (2017-2019) of the U.S. PATH cohort: 9,422 reported no asthma diagnosis and 2,421 reported an asthma diagnosis	Current (in the past 30 days) • Sole e-cig use • Sole cigarette or cigar use • Dual use of cigarettes/cigars and e-cig • Never used tobacco products (control)	Self-report of: • Two respiratory symptoms: wheezing or whistling in the chest or dry cough at night in the past 12 months • Newly diagnosed asthma in the prior 12 months • Asthma-related interference with activities in the prior 30 days	Adjusting for age, sex, race/ethnicity, household income, parental education, and home tobacco rules: • Sole e-cig use or dual use was not associated with respiratory symptoms, newly asthma diagnosis, or asthma-related activities • Sole cigarette/cigar smoking was associated with wheezing/whistling in the chest in the prior 12 months (aOR=1.88, 95% CI=1.22-2.88) • Asthma-related interference with activities most/all of the time (vs. none/little of the time) was lower in sole e-cig users (aOR=0.14, 95% CI=0.02-0.87) than in sole cigarette/cigar smokers, and higher for dual use than for sole e-cig use <u>Limitations:</u> - Potential recall bias - Prevalence of e-cig use in PATH is lower than in other US national cohorts
Pérez et al., 2024 ⁵⁹	17,023 youths aged 12 to 17 years and 7,766 adults that did not have asthma or COPD and never used cigarettes at the baseline (2013-2020) in the US PATH cohort	Current e-cig or ENDS use (in the past 30 days) vs. not current use (control)	• Age of onset of asthma (by parental report in youth; by self-report in adults)	Adjusting for sex, race/ethnicity, parental education, weight, current use any cigarette products, hookah, smokeless tobacco, total years of e-cig use, binge drinking, marijuana use, and SHS at home: • Use of e-cig was not associated with age of asthma onset (aHR=1.79, 95% CI=0.67-4.77) <u>Limitations:</u> - Lack of data on known risk factors for asthma onset such as maternal smoking, diet, allergy, and family history of asthma - The age of asthma onset was estimated by interval-censoring survival method (exact dates not available) - Lack of information of e-cig use frequency, number of puffs, and types of flavoring

ACT=asthma control test, aHR=adjusted hazard ratio, aOR= adjusted odds ratio, aPR= adjusted prevalence ratio, BMI=body mass index, cACT=children asthma control test, CHS=Children Health Study, CI=confidence interval, COPD= chronic obstructive pulmonary disease, e-cig=electronic cigarettes, ENDS= electronic nicotine delivery systems, PATH=Population Assessment of Tobacco and Health study, SHA=secondhand aerosols, SHS=secondhand smoke, YRBS= Youth Risk Behavior Surveillance

Table 4. Use of e-cigarettes and asthma or asthma-related outcomes in adults

Reference	Study population	Exposure(s)	Asthma-related outcome(s)	Key findings and study limitations
Cross-sectional studies				
Pérez et al., 2024 ⁵⁹	7,766 adults that did not have asthma or COPD and never used cigarettes at the baseline (2013-2020) in the US PATH cohort	Current e-cig or ENDS use (in the past 30 days) vs. not current user (control)	Self-report of age of onset of asthma	Adjusting for sex, race/ethnicity, education, weight, current use any cigarette products, hookah, smokeless tobacco, total year of e-cig use, binge drinking, marijuana use, and SHS at home, • Use of e-cig was associated with asthma onset at an earlier age (aHR=3.52, 95% CI=1.24-10.02). <u>Limitations:</u> - Inability to assess temporal relationships - Lack of information on known risk factors for asthma onset - The age of asthma onset was estimated by interval-censoring survival method (exact date not available) - Lack information on e-cig use frequency, number of puffs, and types of flavoring
Delmas et al., 2024 ⁶⁰	136,276 adults aged 18-69 years recruited in 2015-2019 for the Constances cohort, France	Ever (current or former) sole use of disposable or refillable e-cig use vs. never use (control)	Self-report of: • Asthma symptom score in the last 12 months • Symptoms of chronic bronchitis	Adjusting for age, sex, education, BMI, cannabis use and history of chronic respiratory diseases, in never combustible cigarette smokers: • Ever e-cig use was associated with an increased asthma symptom score in both women (aMSR= 1.44, 95% CI= 1.09-1.90) and men (aMSR=1.36, 95% CI=1.01-1.83) • Ever e-cig use was associated with symptoms of chronic bronchitis in women (aPR=1.97, 95% CI= 1.27-3.05), but not in men (aOR=0.97, 95% CI=0.57-1.65) <u>Limitations:</u> - Inability to assess temporal relationships - Potential recall bias - Lack of data on exposure to SHS or cannabis use - Unable to study dose-response relationships

Osei et al., 2019 ⁶¹	402,822 adults ≥18 years in the 2016 and 2017 U.S. BRFSS	Current (now) e-cig use without history of combustible cigarette smoking vs. never use (control)	Self-report of current asthma	Adjusting for age, sex, race, income, education, and BMI: - E-cig use was associated with current asthma (aOR= 1.39, 95% CI= 1.15-1.68) - In current e-cig users, a dose-response association was reported: occasional use: OR=1.31 (95% CI= 1.05-1.62); daily use: OR=1.73 (95% CI= 1.21-2.48) <u>Limitations:</u> - Inability to assess temporal relationships - Potential misclassification of asthma - Lack of data on SHS exposure or cannabis use
Varella et al., 2022 ⁶²	18,079 adults aged 18-44 years in the 2017 U.S. BRFSS	E-cig use (current daily, current some days, former, and never [control])	Self-reported presence of respiratory symptoms (cough, phlegm, or shortness of breath in the past 3 months)	Adjusting for age, sex, race, marital status, employment, education, annual family income, health insurance status, exercise, cigarette smoking, history of asthma or COPD: - Current e-cig use in some days (aOR= 1.49, 95% CI= 1.06-2.11) and former e-cig use (aOR= 1.29, 95% CI=1.07-1.55) were associated with respiratory symptoms <u>Limitations:</u> - Inability to assess temporal relationships - Potential recall bias - Lack of information on exposure to SHS or cannabis use

Eugene et al., 2024 ⁶³	52,731 adults aged ≥ 18 years from the Wave 4 (2016-2018) of the U.S. PATH cohort	Ever e-cig use vs. never use (control)	Self-report of any diagnosis of respiratory illnesses: • Asthma, • Chronic bronchitis • COPD • Emphysema • Other lung or respiratory conditions	Adjusting for age, gender, race/ethnicity, education, insurance status, and use of other tobacco products: • E-cig use was associated with any diagnosis of respiratory illness (aOR=1.32, 95% CI= 1.03-1.69) • Adults ≥ 35 years old (aOR= 1.65, 95 % CI= 1.20-2.26) had greater odds of any respiratory illness than younger adults (aOR=0.95, 95% CI=0.71-1.26) <u>Limitations:</u> - Unmeasured confounders including exposure to SHS, air pollutants, use of cannabis - Potential recall bias - Cannot estimate effects on a specific illness - Inability to assess temporal relationships
Dirisanala et al., 2023 ⁶⁴	178,157 adults aged ≥ 18 years from the 2015-2018 U.S. NHANES	• Exclusive ever e-cig use • Exclusive current cigarette smoking • Dual use of e-cig and traditional cigarettes • Never use either (control)	Self-report current asthma	• Sole e-cig use (aOR=1.47, 95% CI=1.21-1.78), sole cigarette smoking (aOR=1.27, 95% CI=1.15-1.41), and dual use of e-cig and cigarettes (aOR=1.33, 95% CI= 1.19-1.48) were associated with current asthma. <u>Limitations:</u> - The covariates included in the multivariable analysis were not specified - Inability to assess temporal relationships - Potential recall bias - Lack of information on SHS exposures or cannabis use

Adebisi et al., 2025 ⁶⁶	9,000 adults aged ≥ 16 years in the 2017-2019 Scottish Health Survey	Current nicotine use status: • exclusive e-cig use • exclusive cigarette smoking • never use of either product (control)	Self-report of wheezing in the past 12 months.	Adjusting for age, sex, area-level deprivation, chronic illness, doctor-diagnosed COPD or asthma, general health, alcohol consumption, and SHS exposures: • Exclusive e-cig use in former smokers who quit ≤ 5 years ago (aOR=1.61, 95% CI=1.14-2.26) and exclusive cigarette smokers (aOR=2.77, 95% CI=2.34-3.28) was associated with wheezing • Exclusive e-cig use in never smokers, exclusive e-cig use in former smokers who quit > 10 years ago, or exclusive e-cig use in former smokers who quit smoking 6-10 years ago were not associated with wheezing <u>Limitations:</u> - Inability to assess temporal relationships - Potential recall bias - Lack of information on device type, duration, intensity, or use of cannabis, pipes, or cigars
Tanu et al., 2024 ⁶⁷	11,598 adults aged ≥ 18 years in the 2018 U.S. Asthma Call Back Survey linked to the 2018 U.S. BRFSS	• Exclusive ever e-cig use • Exclusive ever cigarette use • Dual use of e-cig and cigarettes (ever) • Never use either product (control)	Self-report of asthma attacks in the previous year and asthma symptoms in the past 30 days among participants with known asthma	Adjusting for age, sex, race/ethnicity, education, BMI, employment, household income, frequency of binge drinking, influenza vaccination, insurance, and comorbidities (history of respiratory diseases, heart attack, and depression): • Exclusive e-cig use (aOR= 0.88; 95% CI= 0.56-1.38), exclusive cigarette use (aOR=0.80; 95% CI= 0.60-1.05), or dual-use (aOR=0.98; 95% CI= 0.68-1.41) were not associated with asthma exacerbations or asthma symptoms. <u>Limitations:</u> - Inability to assess temporal relationships - Potential recall bias - Lack of information on asthma severity, use of asthma medications, or adherence to medications

Human exposure study

Kotoulas et al., 2020 ⁶⁵	50 adults ≥ 18 years (25 healthy smokers and 25 smokers with moderate persistent well-controlled asthma treated with low dose of ICS+LABA), Greece, Year of study was not specified	E-cig use for 5 minutes: 10 puffs with 30 seconds intervals between puffs, 1.0-1.5 mL of e-liquid and 11 mg of nicotine (NOBACCO MLB-MED filter)	Acute outcomes: 15 minutes after vaping: • Lung function • FeNO 30 minutes after vaping: • Inflammatory markers in EBC	After using e-cigs (paired sample test) In smokers with asthma: ↓ %predicted FEV₁/FVC ($p=0.04$) and PEF ($p<0.01$) ↑ FeNO ($p<0.01$) and EBC pH ($p=0.01$) ↑ IL-10 ($p<0.01$) and TNF-α ($p=0.03$) in EBC In healthy smokers: ↓ FeNO ($p<0.01$) <u>Limitations:</u> - Only measured acute effects of e-cigs - Lack a sham control or exposure to different types of device, liquids, or nicotine - Limited generalizability
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ACT=asthma control test, aOR= adjusted odds ratio, aPR= adjusted prevalence ratio, aMSR=adjusted mean score ratio, BMI=body mass index, BRFSS=Behavioral risk factor surveillance system, CI=confidence interval, COPD= chronic obstructive pulmonary disease, EBC=exhaled breath condensate, e-cigarettes=electronic cigarettes, FEV₁= forced expiratory volume in one second, FeNO=Fractional exhaled nitric oxide, FVC= forced vital capacity, ICS=inhaled corticosteroid, LABA=long-acting beta-agonists, NHANES= National Health and Nutrition Examination Survey, PATH= Population Assessment of Tobacco and Health Study, PEF= peak expiratory flow, SHS=secondhand smoking

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